signatures and the verifier’s ability to validate them against a trusted issuer.

- **Queried via APIs.** Other signals are retrieved dynamically at decision time from authoritative systems. These communications typically use TLS for encryption and server authentication, and sometimes employ mutual TLS for stronger assurance.
- Delivered by sidecar agents or message buses. In modern microservice environments, sidecar processes or internal message streams distribute context updates—such as risk scores or revocation notices—to nearby PDPs with minimal latency.

Each approach trades simplicity for greater control. Tokens are self-contained but remain static. API calls are flexible but rely on network connectivity and upstream availability. Streams or agents can deliver updates rapidly, but they need careful deployment and ongoing monitoring.

12.5.2 Protecting signals in transit

The security of signal transmission provides confidence in the signal's authenticity, integrity, and confidentiality. Each system must be able to demonstrate provenance: where the signal originated, who asserted it, and that it hasn’t been altered or exposed during transmission.

For direct, system-to-system calls within ACME, most integrations use TLS encryption for confidentiality and mutual TLS (mTLS) where both endpoints authenticate each other. However, mTLS isn’t always practical—especially when integrating with SaaS platforms or OAuth-based systems that rely on bearer tokens. In those cases, signed tokens provide verifiable provenance.

A signed token includes a cryptographic signature from the issuing system—for example, the posture client on an employee’s laptop or ACME’s customer authentication service. When the Policy Decision Point (PDP) verifies this signature against a known public key, it increases confidence in the token’s authenticity and confirms its contents haven’t been tampered with. This reassurance enables systems to exchange context securely, even if mutual authentication at the transport layer isn’t possible.